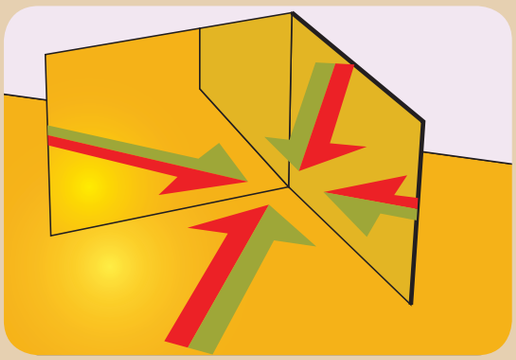
ray are not equal. Hence, the law of reflection is not obeyed in this case and thus the image is not clear. Such a reflection is called 'irregular reflection' or 'diffused reflection'. Example: Reflection of light from a wall.

3.7 Multiple Reflections

Activity 4

Take two plane mirrors and keep them perpendicular to each other. Place an object between them. You can see the images of the object. How many images do you see in the mirrors? You can see three images. How is it possible to have three images with two mirrors?



In the activity given above, you observed that for an object kept in between two plane mirrors, which were inclined to each other, you could see many images. This is because, the 'image' formed by one mirror acts as an 'object' for the other mirror. The image formed by the first mirror acts as an object for the second mirror and the image formed by the second mirror acts as an object for the first mirror. Thus, we have three images of a single body. This is known as multiple reflection. This type of reflections can be seen in show rooms and saloons.

The number of images formed, depends on the angle of inclination of the mirrors. If the angle between the two mirrors is a factor of 360°, then the total number of reflections is finite. If θ (Theta) is the angle of inclination of the plane mirrors, the number of images formed is equal to $\frac{360^\circ}{\theta} - 1$. As you decrease this angle, the number of images formed increases. When they are parallel to each other, the number of images formed becomes infinite.

Problem 3

If two plane mirrors are inclined to each other at an angle of 90°, find the number of images formed.

Solution

Angle of inclination = 90°

$$\text{Number of images formed} = \frac{360^\circ}{\theta} - 1$$

$$= \frac{360^\circ}{90^\circ} - 1 = 4 - 1 = 3$$

3.7.1 Kaleidoscope

It is a device which functions on the principle of multiple reflection of light, to produce numerous patterns of images. It has two or more mirrors inclined to each other. It can be designed from inexpensive materials. The colourful image patterns formed by this will be pleasing to you. This instrument is used as a toy for children.



Figure 3.9 Kaleidoscope

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